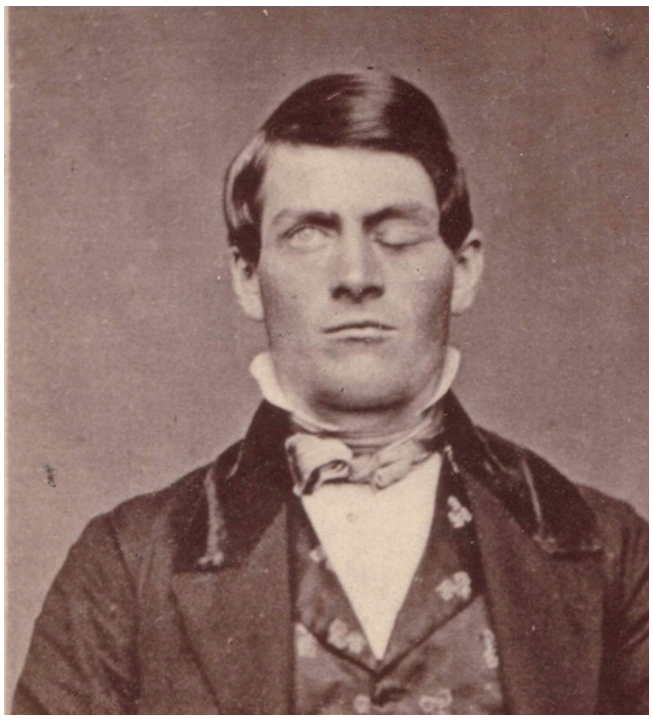


PHINEAS GAGE

By: Devika Mathur



ABOUT PHINEAS GAGE

. Born in born July 1823, New Hampshire, U.S.—died May 1860, California

. Started working with contractors who were associated with the Rutland and Burlington Railroad.

. On September 13th, 1848 he suffered a terrible railroad accident while working with gunpowder.

. A six Kg rod had flown back against his skull and went in through his Left Frontal Lobe And twenty five meters behind him, despite this, He survived.

. After this accident, he seemed to be another person in the same body. He became disrespectful, unreliable, childish and restless, a complete swap from his former self.

THE DETAILS OF THE INCIDENT

. Gage was given a task to direct a Rock-splitting work group while preparing a bed for Rutland & Burlington Railroad south of Cavendish, Vermont, USA

. The gunpowder was pressed into the hole with a bar and exploded with friction, resulting in a blast which made the meter long bar shoot through his skull.

. After it entered his left side and shot right back out, he woke up after some convulsions and walked to the ox cart back to his quarters with little help.

. After Dr. Edward H. Williams was called, gage was treated with medical care. He had lost a lot of blood and the wound would constantly get affected.

. He was anemiac and fell into a Semicomatose state and developed a fungal infection that needed surgery. afterwards, he was walking around the city win mid-november.

THE AFTERMATH AND CHANGES

. Although Gage didn't seem to have significant memory, cognition and strength, he seemed to be a different person.

. He became rude, childish and indecisive; his mind had changed for the worse and his colleagues described his as 'not Gage'.

. He was fired for indiscipline and started his new life as a circus attraction. It is assumed that he convert back to his old self after four years, as proved by his work as long haul driver in Chile; A job that requires planning and focus.

. After a while, he died on May 21st, 1861, which was twelve years after his accident due to an epileptic seizure closely related to his brain injury.

THE FRONTAL BRAIN AND PREFRONTAL CORTEX

. Located at the front part of the brain behind the forehead; involved in problem-solving, planning and focus.

. Last part of the brain to finish myelination, a process important for neuron signaling.

FUNCTIONS

. Working memory - short term information.

. Attention - staying focused despite distractions.

. Decision making - efficient planning and thinking of choices

. Goal Orientation - helps you stay on track to manage impulse from past experiences

. Emotions - helps us to empathise and understand feelings.

WHAT HAPPENED WITH GAGE?

-> The injury to his frontbal brain, specifically the prefrontal cortex had played a significant role in his personality change.

-> The heavy metal bar destroyed soft tissues, facial arteries, and blood vessels inside both his left cheek and his cerebral cortex (outermost layer of folded gray matter).

-> Gage's case had a significant impact on the future of neuropsychiatry and Neuroscience. He was a base for further research on the prefrontal cortex and its impact on human personality and daily function.

-> Ventromedial pfc- loss of ability to decide.

-> Orbitofrontal cortex - inability to learn from past blunders.

-> Dorsolateral PFC - inability to organise or execute plans.

-> Anterior Singulate - unaware of mistakes

-> Right Frontal Lobe
- lack of social empathy and cognition

THANK YOU!

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